

1. Параметры VM и модель GPU

CPU, ядра, потоки

devchatay@epds73cmval2ppf6qhpm:~\$ lscpu

Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 40 bits physical, 57 bits virtual
Byte Order: Little Endian
CPU(s): 8
On-line CPU(s) list: 0-7
Vendor ID: GenuineIntel
Model name: Intel Xeon Processor (Icelake)
CPU family: 6
Model: 106
Thread(s) per core: 2
Core(s) per socket: 4
Socket(s): 1
Stepping: 0
BogoMIPS: 4000.00
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fx
sr sse sse2 ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology nonst
op_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic mov
be popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
fetch cpuid_fault invpcid_single ssbd ibrs ibpb stibp ibrs_enhanced fsgsbase bmi1 avx2
smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb av
x512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 wbnoinvd arat avx512vbmi umip
pku ospke avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpoperndq
la57 rdpid fsrm md_clear arch_capabilities

Virtualization features:

Hypervisor vendor: KVM

Virtualization type: full

Caches (sum of all):

L1d: 256 KiB (8 instances)

L1i: 256 KiB (8 instances)

L2: 16 MiB (4 instances)

L3: 16 MiB (1 instance)

NUMA:

NUMA node(s): 1

NUMA node0 CPU(s): 0-7

Vulnerabilities:

Gather data sampling: Not affected

Indirect target selection: Mitigation; Aligned branch/return thunks

Itlb multihit: Not affected

L1tf: Not affected

Mds: Not affected

Meltdown: Not affected

Mmio stale data: Mitigation; Clear CPU buffers; SMT Host state unknown

Reg file data sampling: Not affected

Retbleed: Not affected

Spec rstack overflow: Not affected

Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl and seccomp

Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization

Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBR SB-eIBRS SW sequence; BHI

SW loop, KVM SW loop

Srbds: Not affected

Tsa: Not affected

Tsx async abort: Not affected

RAM

devchatay@epds73cmval2ppf6qhpm:~\$ free -h

	total	used	free	shared	buff/cache	available
Mem:	31Gi	3.2Gi	13Gi	106Mi	14Gi	27Gi

Swap: 0B 0B 0B

Модель GPU

```
devchatay@epds73cmval2ppf6qhpm:~$ lspci | grep -i nvidia
8b:00.0 3D controller: NVIDIA Corporation TU104GL [Tesla T4] (rev a1)
```

Тип VM

```
devchatay@epds73cmval2ppf6qhpm:~$ sudo dmidecode -t system
# dmidecode 3.3
Getting SMBIOS data from sysfs.
SMBIOS 2.8 present.
Handle 0x0100, DMI type 1, 27 bytes
System Information
    Manufacturer: Yandex
    Product Name: xeon-gold-6338
    Version: pc-q35-yc-5.0
    Serial Number: YC-epds73cmval2ppf6qhpm
    UUID: 23000007-65bc-38d9-6faa-a2ce5e6d4736
    Wake-up Type: Power Switch
    SKU Number: Not Specified
    Family: Yandex Cloud
Handle 0x2000, DMI type 32, 11 bytes
System Boot Information
    Status: No errors detected
```

2. Версии Ubuntu, Kubernetes, GPU Operator, драйвера NVIDIA и CUDA runtime

Версия Ubuntu

```
devchatay@epds73cmval2ppf6qhpm:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 22.04.5 LTS
Release: 22.04
Codename: jammy
```

Версия Kubernetes

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl version --client
Client Version: v1.34.8
Kustomize Version: v5.7.1
```

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl version
Client Version: v1.34.8
Kustomize Version: v5.7.1
Server Version: v1.34.8
```

Версия NVIDIA GPU Operator

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl get deployment gpu-operator -n gpu-operator -o jsonpath='{.spec.template.spec.containers[0].image}'
```

nvcr.io/nvidia/gpu-operator:v25.10.1

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl get pods -n gpu-operator -l app=gpu-operator -o jsonpath='{.items[0].spec.containers[0].image}'
nvcr.io/nvidia/gpu-operator:v25.10.1
```

Версия драйвера NVIDIA

```
devchatay@epds73cmval2ppf6qhpm:~$ cat /proc/driver/nvidia/version | head -n 1
NVRM version: NVIDIA UNIX Open Kernel Module for x86_64 580.105.08 Release Build (dvs-builder@U22-I3-B10-02-5)
Wed Oct 29 22:29:53 UTC 2025
```

Версия CUDA Runtime

```
devchatay@epds73cmval2ppf6qhpm:~$ nvcc --version
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2021 NVIDIA Corporation
Built on Thu_Nov_18_09:45:30_PST_2021
Cuda compilation tools, release 11.5, V11.5.119
Build cuda_11.5.r11.5/compiler.30672275_0
```

3. Скриншот или текстовый вывод kubectl get nodes -o wide

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl get nodes -o wide
NAME                                STATUS    ROLES    AGE     VERSION    INTERNAL-IP    EXTERNAL-IP    OS-IMAGE             KERNEL-VERSION    CONTAINER-RUNTIME
epds73cmval2ppf6qhpm               Ready     control-plane  5d22h    v1.34.8    10.129.0.25    <none>         Ubuntu 22.04.5 LTS   5.15.0-157-generic containerd://2.2.1
```

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl get nodes -o wide;
NAME                                STATUS    ROLES    AGE     VERSION    INTERNAL-IP    EXTERNAL-IP    OS-IMAGE             KERNEL-VERSION    CONTAINER-RUNTIME
epds73cmval2ppf6qhpm               Ready     control-plane  5d21h    v1.34.8    10.129.0.25    <none>         Ubuntu 22.04.5 LTS   5.15.0-157-generic
containerd://2.2.1
```

4. Скриншот или текстовый вывод kubectl get pods -A

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl get pods -A
NAMESPACE   NAME                                READY   STATUS    RESTARTS   AGE
gpu-lab     torch-training-exclusive-wdwbz     0/1     Completed 0          27m
gpu-operator gpu-feature-discovery-dxjbl         2/2     Running   0          4d19h
gpu-operator gpu-operator-7569f8b499-5ck8q       1/1     Running   9 (4h8m ago) 4d19h
gpu-operator gpu-operator-node-feature-discovery-gc-55ffc49ccc-btq9c 1/1     Running   0          4d19h
gpu-operator gpu-operator-node-feature-discovery-master-6b5787f695-pw6z6 1/1     Running   6 (4h8m ago) 4d19h
gpu-operator gpu-operator-node-feature-discovery-worker-x9m5g       1/1     Running   0          4d19h
gpu-operator nvidia-container-toolkit-daemonset-xzb7f 1/1     Running   0          4d19h
gpu-operator nvidia-cuda-validator-cbbch         0/1     Completed 0          4d19h
gpu-operator nvidia-dcgm-exporter-qtfr           1/1     Running   0          14h
gpu-operator nvidia-device-plugin-daemonset-qhpfx 2/2     Running   0          31m
gpu-operator nvidia-driver-daemonset-gvv8v       1/1     Running   0          4d19h
gpu-operator nvidia-operator-validator-cnkmn    1/1     Running   0          4d19h
kube-flannel kube-flannel-ds-mgb79               1/1     Running   0          5d21h
kube-system coredns-66bc5c9577-87hsd           1/1     Running   0          5d21h
kube-system coredns-66bc5c9577-f5xsv           1/1     Running   0          5d21h
```

kube-system	etcd-epds73cmval2ppf6qhpm	1/1	Running	0	5d21h
kube-system	kube-apiserver-epds73cmval2ppf6qhpm	1/1	Running	0	5d21h
kube-system	kube-controller-manager-epds73cmval2ppf6qhpm	1/1	Running	8 (4h21m ago)	5d21h
kube-system	kube-proxy-lvjvx	1/1	Running	0	5d21h
kube-system	kube-scheduler-epds73cmval2ppf6qhpm	1/1	Running	8 (4h8m ago)	5d21h
monitoring	alertmanager-monitoring-kube-prometheus-alertmanager-0	2/2	Running	0	4d18h
monitoring	monitoring-grafana-6b866c5c-jzl64	3/3	Running	0	4d18h
monitoring	monitoring-kube-prometheus-operator-554ffdbcd5-9p1ls	1/1	Running	0	28m
monitoring	monitoring-kube-state-metrics-7f7f8474d9-vnk8f	1/1	Running	1 (5h ago)	4d18h
monitoring	monitoring-prometheus-node-exporter-gnjzc	1/1	Running	0	4d18h
monitoring	prometheus-monitoring-kube-prometheus-prometheus-0	2/2	Running	0	14h

5. Вывод nvidia-smi из pod;

Манифест для запуска пода и исполнения nvidia-smi

```
devchatay@epds73cmval2ppf6qhpm:~$ cat > ~/manifests/torch-training-testnvidiasmi.yaml << 'EOF'
apiVersion: batch/v1
kind: Job
metadata:
  name: nvidia-smi-test
  namespace: gpu-lab
spec:
  ttlSecondsAfterFinished: 3600
  backoffLimit: 0
  template:
    metadata:
      labels:
        app: nvidia-smi-test
    spec:
      restartPolicy: Never
      containers:
        - name: smi
          image: pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime
          imagePullPolicy: IfNotPresent
          command: ["/bin/bash", "-c"]
          args:
            - |
              echo "=== NVIDIA-SMI ==="
              nvidia-smi
              echo ""
              echo "=== CUDA Runtime (nvcc) ==="
              nvcc --version 2>/dev/null || echo "nvcc not found in this image"
EOF      nvidia.com/gpu: "1" n: ', torch.backends.cudnn.version())" 2>/dev/null || echo "PyTorch not available".cuda.is_available());
print('CUDA version (compiled with):', torch.
cat > ~/manifests/torch-training-testnvidiasmi.yaml << 'EOF'
apiVersion: batch/v1
kind: Job
metadata:
  name: nvidia-smi-test
  namespace: gpu-lab
spec:
  ttlSecondsAfterFinished: 3600
  backoffLimit: 0
  template:
    metadata:
      labels:
        app: nvidia-smi-test
    spec:
```

```

restartPolicy: Never
containers:
- name: smi
  image: pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime
  imagePullPolicy: IfNotPresent
  command: ["/bin/bash", "-c"]
  args:
  - |
    echo "=== NVIDIA-SMI ==="
    nvidia-smi
    echo ""
    echo "=== CUDA Runtime (nvcc) ==="
    nvcc --version 2>/dev/null || echo "nvcc not found in this image"
    echo ""
    echo "=== PyTorch CUDA info ==="
    python -c "import torch; print('PyTorch version:', torch.__version__); print('CUDA available:', torch.cuda.is_available());
    print('CUDA version (compiled with):', torch.version.cuda); print('cuDNN version:', torch.backends.cudnn.version())" 2>/dev/null || echo
    "PyTorch not available"
  resources:
    limits:
      nvidia.com/gpu: "1"
    requests:
      nvidia.com/gpu: "1"
EOF

```

Исполнение манифеста и результат в логах:

```

devchatay@epds73cmval2ppf6qhpm:~$ kubectl apply -f ~/manifests/torch-training-testnvidiasmi.yaml
job.batch/nvidia-smi-test created
devchatay@epds73cmval2ppf6qhpm:~$ kubectl logs -n gpu-lab job/nvidia-smi-test
=== NVIDIA-SMI ===
Sun May 31 12:34:31 2026
+-----+
| NVIDIA-SMI 580.105.08                  Driver Version: 580.105.08          CUDA Version: 13.0                  |
+-----+-----+-----+-----+-----+-----+-----+-----+
| GPU   Name      Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC | | |
| Fan   Temp      Perf             Pwr:Usage/Cap     |           |         | GPU-Util  Compute M. |
|              |              Pwr:Usage/Cap     |           |         | GPU-Util  Compute M. |
|====+=====+====+=====+=====+=====+=====+=====+
|    0   Tesla T4               On   | 00000000:8B:00:00 Off |             |      0 | |
| N/A   35C      P8             14W / 70W | 0MiB / 15360MiB |             | 0%    | Default |
|              |              Pwr:Usage/Cap     |           |         | GPU-Util  Compute M. |
|              |              Pwr:Usage/Cap     |           |         | GPU-Util  Compute M. |
+-----+-----+-----+-----+-----+-----+-----+
+-----+
| Processes:                               |
|  GPU   GI    CI          PID    Type   Process name                      GPU Memory |
|      ID   ID                 |           |         |                               | Usage    |
+-----+-----+-----+-----+-----+-----+
| No running processes found              |
+-----+

=== CUDA Runtime (nvcc) ===
nvcc not found in this image

=== PyTorch CUDA info ===
PyTorch version: 2.5.1+cu124
CUDA available: True
CUDA version (compiled with): 12.4
cuDNN version: 90100
devchatay@epds73cmval2ppf6qhpm:~$ █

```

Выполнялось в состоянии, когда другие процессы (другие нагрузки на GPU, обучения) не исполнялись

6. Результаты benchmark для exclusive, time-slicing и, если получилось, MPS;

В качестве задачи сначала было взято дообучение (fine-tuning) LLM. Пример exclusive yaml-файла:

```
cat > ~/manifests/torch-training-shared.yaml << 'EOF'
apiVersion: batch/v1
kind: Job
metadata:
  name: torch-training-shared
  namespace: gpu-lab
spec:
  ttlSecondsAfterFinished: 3600
  backoffLimit: 0
  template:
    metadata:
      labels:
        app: torch-training
        mode: shared
    spec:
      restartPolicy: Never
      containers:
        - name: trainer
          image: pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime
          imagePullPolicy: IfNotPresent
          command:
            - /bin/bash
            - -c
            - |
              set -e
              echo "Installing dependencies..."
              pip install --no-cache-dir --quiet transformers==4.40.0 accelerate==0.29.0 datasets==2.19.0

              echo "Starting training..."
              python -c "
              import os, time, json, torch
              from transformers import AutoTokenizer, AutoModelForCausalLM, TrainingArguments, Trainer, TrainerCallback
              import torch.utils.data

              step_times, peak_mem_gb = [], 0
              last_t = None

              def log_gpu_info():
                  if torch.cuda.is_available():
                      name = torch.cuda.get_device_name(0)
                      total_mem = torch.cuda.get_device_properties(0).total_memory / 1024**3
                      print(f'GPU: {name}')
                      print(f'TOTAL_GPU_MEMORY_GB: {total_mem:.2f}')
                      return name, total_mem
                  return 'Unknown', 0

              class MetricsCallback(TrainerCallback):
                  def on_step_begin(self, args, state, control, **kwargs):
                      global last_t
                      last_t = time.time()
                      torch.cuda.reset_peak_memory_stats()
                  def on_step_end(self, args, state, control, **kwargs):
                      global peak_mem_gb
                      if last_t:
                          step_times.append(time.time() - last_t)
                          peak_mem_gb = max(peak_mem_gb, torch.cuda.max_memory_allocated() / 1024**3)
                      if state.global_step % 5 == 0:
                          print(f'Step {state.global_step}: time={step_times[-1]:.3f}s, peak_mem={peak_mem_gb:.2f}GB')
                  def on_train_end(self, args, state, control, **kwargs):
                      if step_times:
                          avg, mn, mx = sum(step_times)/len(step_times), min(step_times), max(step_times)
                          print(f'\n * "\n')
                          print(f'avg_step_sec: {avg:.4f}')
                          print(f'min_step_sec: {mn:.4f}')
                          print(f'max_step_sec: {mx:.4f}')
                          print(f'peak_cuda_mem_gb: {peak_mem_gb:.2f}')
                          print(f'METRICS_JSON: {json.dumps({"avg_step_sec":round(avg,4),"min_step_sec":round(mn,4),"max_step_sec":round(mx,4),"peak_cuda_mem_gb":round(peak_mem_gb,2)})}')
                          print(f'=\n\n')

              gpu_name, total_mem = log_gpu_info()
              model_name = 'Owen/Owen2.5-7.5B-Instruct'
              hf_token = os.getenv('HF_TOKEN', '')
```

```

        peak_mem_gb = max(peak_mem_gb, torch.cuda.max_memory_allocated() / 1024**3)
    if state.global_step % 5 == 0:
        print(f'Step {state.global_step}: time={step_times[-1]:.3f}s, peak_mem={peak_mem_gb:.2f}GB')
    def on_train_end(self, args, state, control, **kwargs):
        if step_times:
            avg, mn, mx = sum(step_times)/len(step_times), min(step_times), max(step_times)
            print('\n' + '='*60)
            print(f'avg_step_sec: {avg:.4f}')
            print(f'min_step_sec: {mn:.4f}')
            print(f'max_step_sec: {mx:.4f}')
            print(f'peak_cuda_mem_gb: {peak_mem_gb:.2f}')
            print(f'METRICS_350N: {json.dumps({"avg_step_sec":round(avg,4),"min_step_sec":round(mn,4),"max_step_sec":round(mx,4),"peak_cuda_mem_gb":round(peak_mem_gb,2)})}')
            print('='*60 + '\n')

    gpu_name, total_mem = log_gpu_info()
    model_name = 'Qwen/Qwen2.5-0.5B-Instruct'
    hf_token = os.getenv('HF_TOKEN', '')

    tokenizer = AutoTokenizer.from_pretrained(model_name, token=hf_token)
    model = AutoModelForCausalLM.from_pretrained(model_name, token=hf_token).cuda()

    train_texts = ['MOCK DATASET TEXT FOR MODEL TRAINING' * 5] * 500
    encodings = tokenizer(train_texts, truncation=True, padding=True, max_length=64)

    class SimpleDataset(torch.utils.data.Dataset):
        def __init__(self, enc): self.enc = enc
        def __getitem__(self, idx):
            item = (k, torch.tensor(v[idx]) for k, v in self.enc.items())
            item['labels'] = item['input_ids'].clone()
            item['labels'][item['labels'] == tokenizer.pad_token_id] = -100
            return item
        def __len__(self): return len(self.enc['input_ids'])

    training_args = TrainingArguments(
        output_dir='./results',
        num_train_epochs=3,
        per_device_train_batch_size=1,
        save_steps=10000,
        logging_steps=5,
        bf16=torch.cuda.is_bf16_supported(),
        report_to='none',
        gradient_accumulation_steps=2,
    )

    trainer = Trainer(
        model=model, args=training_args, train_dataset=SimpleDataset(encodings),
        callbacks=[MetricsCallback()]
    )

    print('Starting training...')
    t0 = time.time()
    trainer.train()
    print(f'Done in {time.time()-t0:.2f} sec')

env:
- name: HF_TOKEN
  valueFrom:
    secretKeyRef:
      name: hf-token-secret
      key: HF_TOKEN
  optional: true
resources:
  limits:
    cpu: "1"
    memory: 2Gi
  requests:
    cpu: "1"
    memory: 1Gi
  nvidia.com/gpu/shared: "1"
EOF

```

cat > ~/manifests/torch-training-exclusive.yaml << 'EOF'

apiVersion: batch/v1

kind: Job

metadata:

name: torch-training-exclusive

namespace: gpu-lab

spec:

ttlSecondsAfterFinished: 3600

backoffLimit: 0

template:

metadata:

labels:

app: torch-training

mode: exclusive

spec:

restartPolicy: Never

containers:

- name: trainer

image: pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime

imagePullPolicy: IfNotPresent

command:

- /bin/bash

- -c

- |

set -e

echo "Installing dependencies..."

pip install --no-cache-dir --quiet transformers==4.40.0 accelerate==0.29.0

echo "Starting training..."

python -c "

import os, time, json, torch, sys, gc

from transformers import AutoTokenizer, AutoModelForCausalLM, TrainingArguments, Trainer, TrainerCallback

import torch.utils.data

step_times, peak_mem_gb = [], 0

last_t = None

def log_gpu_info():

if torch.cuda.is_available():

name = torch.cuda.get_device_name(0)

total_mem = torch.cuda.get_device_properties(0).total_memory / 1024**3

print(f'GPU: {name}')

print(f'TOTAL_GPU_MEMORY_GB: {total_mem:.2f}')

return name, total_mem

return 'Unknown', 0

class MetricsCallback(TrainerCallback):

def on_step_begin(self, args, state, control, **kwargs):

global last_t

last_t = time.time()

torch.cuda.reset_peak_memory_stats()

def on_step_end(self, args, state, control, **kwargs):


```

global peak_mem_gb

if last_t:

    step_times.append(time.time() - last_t)

    peak_mem_gb = max(peak_mem_gb, torch.cuda.max_memory_allocated() / 1024**3)

if state.global_step % 2 == 0:

    print(f'Step {state.global_step}: time={step_times[-1]:.3f}s, peak_mem={peak_mem_gb:.2f}GB')

    sys.stdout.flush()

# Очистка кэша после каждого шага для экономии памяти

if state.global_step % 5 == 0:

    torch.cuda.empty_cache()

    gc.collect()

def on_train_end(self, args, state, control, **kwargs):

    if step_times:

        avg, mn, mx = sum(step_times)/len(step_times), min(step_times), max(step_times)

        print('\n' + '='*60)

        print(f'avg_step_sec: {avg:.4f}')

        print(f'min_step_sec: {mn:.4f}')

        print(f'max_step_sec: {mx:.4f}')

        print(f'peak_cuda_mem_gb: {peak_mem_gb:.2f}')

        print(f'METRICS_JSON:
{json.dumps({"avg_step_sec":round(avg,4),"min_step_sec":round(mn,4),"max_step_sec":round(mx,4),"peak_cuda_mem_gb":round(peak_mem_gb,2)})}')

        print('='*60 + '\n')

        sys.stdout.flush()

gpu_name, total_mem = log_gpu_info()

model_name = 'Qwen/Qwen2.5-0.5B-Instruct'

hf_token = os.getenv('HF_TOKEN', '')

tokenizer = AutoTokenizer.from_pretrained(model_name, token=hf_token)

model = AutoModelForCausalLM.from_pretrained(model_name, token=hf_token).cuda()

train_texts = ['GPU lab training text' * 3] * 100

```

```
encodings = tokenizer(train_texts, truncation=True, padding=True, max_length=32)
```

```
class SimpleDataset(torch.utils.data.Dataset):

    def __init__(self, enc): self.enc = enc

    def __getitem__(self, idx):

        item = {k: torch.tensor(v[idx]) for k, v in self.enc.items()}

        item['labels'] = item['input_ids'].clone()

        item['labels'][item['labels'] == tokenizer.pad_token_id] = -100

        return item

    def __len__(self): return len(self.enc['input_ids'])
```

```
training_args = TrainingArguments(

    output_dir='./results',

    num_train_epochs=1,

    per_device_train_batch_size=1,

    save_steps=10000,

    logging_steps=2,

    bf16=torch.cuda.is_bf16_supported(),

    report_to='none',

    optim='adafactor',

    gradient_checkpointing=True,

    dataloader_num_workers=0,

)
```

```
trainer = Trainer(

    model=model, args=training_args, train_dataset=SimpleDataset(encodings),

    callbacks=[MetricsCallback()]

)
```

```
print('Starting training...')
```

```
print('TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!')
```

```
sys.stdout.flush()
```

```
t0 = time.time()
```

```
trainer.train()

print(f'Done in {time.time()-t0:.2f} sec')

"

env:

- name: HF_TOKEN

valueFrom:

secretKeyRef:

name: hf-token-secret

key: HF_TOKEN

optional: true

resources:

limits:

cpu: "2"

memory: 4Gi

nvidia.com/gpu: "1"

requests:

cpu: "1"

memory: 2Gi

nvidia.com/gpu: "1"

EOF
```

Но возникли ограничения с shared - 4 пода не влезали по памяти на 1 Tesla T4.

Была взята другая задача - обучение на задачу классификации. Сеть состоит из линейных слоев, дропаутов и активаций. Чтобы сетка оправдывал использование GPU и был CPU тяжело исполняемым, взяла большой hidden_dim, batch_size. Таким образом увеличиваем нагрузку на память и делаем больше параллельных вычислений, на что как раз таки рассчитано GPU. Чтобы задача исполнялась не слишком быстро, взято много данных и поставлено большое количество эпох.

Файлы итоговых манифестов:

EXCLUSIVE

```
devchatay@epds73cmval2ppf6qhpm:~/manifests$ cat torch-training-exclusive.yaml
```

```
apiVersion: batch/v1
```

```
kind: Job
```

```
metadata:
```

```
  name: torch-training-exclusive
```

```
  namespace: gpu-lab
```

```
spec:
```

```
  ttlSecondsAfterFinished: 3600
```

```
  backoffLimit: 0
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: torch-training
```

```
        mode: exclusive
```

```
    spec:
```

```
      restartPolicy: Never
```

```
      containers:
```

```
        - name: trainer
```

```
          image: pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime
```

```
          imagePullPolicy: IfNotPresent
```

```
          command:
```

```
            - /bin/bash
```

```
            - -c
```

```
            - |
```

```
              set -e
```

```
              echo "Installing dependencies..."
```

```
              pip install --no-cache-dir --quiet transformers==4.40.0 accelerate==0.29.0
```

```
              echo "Starting training..."
```

```
              python -c "
```

```
import os, time, json, torch, sys, gc
```

```
import torch.nn as nn
```

```
import torch.utils.data
```

```
from transformers import TrainerCallback
```

```
step_times, peak_mem_gb = [], 0
```

```
last_t = None
```

```
def log_gpu_info():
```

```
    if torch.cuda.is_available():
```

```
        name = torch.cuda.get_device_name(0)
```

```
        total_mem = torch.cuda.get_device_properties(0).total_memory / 1024**3
```

```
        print(f'GPU: {name}')
```

```
        print(f'TOTAL_GPU_MEMORY_GB: {total_mem:.2f}')
```

```
        return name, total_mem
```

```
    return 'Unknown', 0
```

```
class MetricsCallback(TrainerCallback):
```

```
    def on_step_begin(self, args, state, control, **kwargs):
```

```
        global last_t
```

```
        last_t = time.time()
```

```
        torch.cuda.reset_peak_memory_stats()
```

```
    def on_step_end(self, args, state, control, **kwargs):
```

```
        global peak_mem_gb
```

```
        if last_t:
```

```
            step_times.append(time.time() - last_t)
```

```
            peak_mem_gb = max(peak_mem_gb, torch.cuda.max_memory_allocated() / 1024**3)
```

```
        if state.global_step % 2 == 0:
```

```
            print(f'Step {state.global_step}: time={step_times[-1]:.3f}s, peak_mem={peak_mem_gb:.2f} GB')
```

```
            sys.stdout.flush()
```

```
        if state.global_step % 5 == 0:
```

```
            torch.cuda.empty_cache()
```

```
            gc.collect()
```

```
    def on_train_end(self, args, state, control, **kwargs):
```

```
        if step_times:
```

```
            avg, mn, mx = sum(step_times)/len(step_times), min(step_times), max(step_times)
```

```
            print("\n" + '='*60)
```

```
            print(f'avg_step_sec: {avg:.4f}')
```

```
            print(f'min_step_sec: {mn:.4f}')
```

```

        print(f'max_step_sec: {mx:.4f}')
        print(f'peak_cuda_mem_gb: {peak_mem_gb:.2f}')
        print(f'METRICS_JSON:
{json.dumps({"avg_step_sec":round(avg,4),"min_step_sec":round(mn,4),"max_step_sec":round(mx,4),"peak_cuda_mem_gb":round(peak_mem_gb,2)})})')
        print('='*60 + '\n')
        sys.stdout.flush()
class MockState:
    global_step = 0
class MockArgs: pass
class MockControl: pass
gpu_name, total_mem = log_gpu_info()

# Синтетический датасет для классификации
N_SAMPLES = 100000
INPUT_DIM = 256
HIDDEN_DIM = 4096 # Большие слои для нагрузки на память
NUM_CLASSES = 10
BATCH_SIZE = 16384 # большой батч для нагрузки на память
NUM_EPOCHS = 50
torch.manual_seed(42)
X = torch.randn(N_SAMPLES, INPUT_DIM).float()
y = torch.argmax(X[:, :NUM_CLASSES] + torch.randn(N_SAMPLES, NUM_CLASSES) * 0.3, dim=1)
class SyntheticDataset(torch.utils.data.Dataset):
    def __init__(self, X, y):
        self.X = X
        self.y = y
    def __getitem__(self, idx):
        return {'input_ids': self.X[idx], 'labels': self.y[idx]}
    def __len__(self): return len(self.X)
model = nn.Sequential(
    nn.Linear(INPUT_DIM, HIDDEN_DIM),
    nn.ReLU(),
    nn.Dropout(0.1),
    nn.Linear(HIDDEN_DIM, HIDDEN_DIM),
    nn.ReLU(),
    nn.Dropout(0.1),
    nn.Linear(HIDDEN_DIM, HIDDEN_DIM),
    nn.ReLU(),
    nn.Dropout(0.1),
    nn.Linear(HIDDEN_DIM, HIDDEN_DIM // 2),
    nn.ReLU(),
    nn.Linear(HIDDEN_DIM // 2, NUM_CLASSES)
).cuda()
device = next(model.parameters()).device
print(device)
total_params = sum(p.numel() for p in model.parameters())
print(f'Всего параметров: {total_params}')
trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)
print(f'Обучаемых параметров: {trainable_params}')
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
dataset = SyntheticDataset(X, y)
metrics = MetricsCallback()
state = MockState()
args = MockArgs()
control = MockControl()
print('Starting training...')
print('TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!')
sys.stdout.flush()
t0 = time.time()
for epoch in range(NUM_EPOCHS):
    model.train()
    for i in range(0, len(dataset), BATCH_SIZE):
        metrics.on_step_begin(args, state, control)

```

```

        end_idx = min(i + BATCH_SIZE, len(dataset))
        batch_X = dataset.X[i:end_idx].cuda()
        batch_y = dataset.y[i:end_idx].cuda()

        optimizer.zero_grad()
        output = model(batch_X)
        loss = criterion(output, batch_y)
        loss.backward()
        optimizer.step()

        state.global_step += 1
        metrics.on_step_end(args, state, control)

    if epoch % 5 == 0:
        torch.cuda.empty_cache()
        gc.collect()
        metrics.on_train_end(args, state, control)
        print(f'Done in {time.time()-t0:.2f} sec')
    "
env:
- name: HF_TOKEN
  valueFrom:
    secretKeyRef:
      name: hf-token-secret
      key: HF_TOKEN
      optional: true
resources:
  limits:
    cpu: "2"
    memory: 4Gi
    nvidia.com/gpu: "1"
  requests:
    cpu: "1"
    memory: 2Gi
    nvidia.com/gpu: "1"

```

SHARED

`devchatay@epds73cmval2ppf6qhpm:~/manifests$ cat torch-training-shared.yaml`

```

apiVersion: batch/v1
kind: Job
metadata:
  name: torch-training-shared
  namespace: gpu-lab
spec:
  ttlSecondsAfterFinished: 3600
  backoffLimit: 0
  template:
    metadata:
      labels:
        app: torch-training
        mode: shared
    spec:
      restartPolicy: Never
      containers:
        - name: trainer
          image: pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime
          imagePullPolicy: IfNotPresent
          command:
            - /bin/bash
            - -c

```

```

- |
set -e
echo "Installing dependencies..."
pip install --no-cache-dir --quiet transformers==4.40.0 accelerate==0.29.0

echo "Starting training..."
python -c "
import os, time, json, torch, sys, gc
import torch.nn as nn
import torch.utils.data
from transformers import TrainerCallback
step_times, peak_mem_gb = [], 0
last_t = None
def log_gpu_info():
    if torch.cuda.is_available():
        name = torch.cuda.get_device_name(0)
        total_mem = torch.cuda.get_device_properties(0).total_memory / 1024**3
        print(f'GPU: {name}')
        print(f'TOTAL_GPU_MEMORY_GB: {total_mem:.2f}')
        return name, total_mem
    return 'Unknown', 0
class MetricsCallback(TrainerCallback):
    def on_step_begin(self, args, state, control, **kwargs):
        global last_t
        last_t = time.time()
        torch.cuda.reset_peak_memory_stats()
    def on_step_end(self, args, state, control, **kwargs):
        global peak_mem_gb
        if last_t:
            step_times.append(time.time() - last_t)
            peak_mem_gb = max(peak_mem_gb, torch.cuda.max_memory_allocated() / 1024**3)
        if state.global_step % 2 == 0:
            print(f'Step {state.global_step}: time={step_times[-1]:.3f}s, peak_mem={peak_mem_gb:.2f}GB')
            sys.stdout.flush()
        if state.global_step % 5 == 0:
            torch.cuda.empty_cache()
            gc.collect()
    def on_train_end(self, args, state, control, **kwargs):
        if step_times:
            avg, mn, mx = sum(step_times)/len(step_times), min(step_times), max(step_times)
            print("\n" + '='*60)
            print(f'avg_step_sec: {avg:.4f}')
            print(f'min_step_sec: {mn:.4f}')
            print(f'max_step_sec: {mx:.4f}')
            print(f'peak_cuda_mem_gb: {peak_mem_gb:.2f}')
            print(f'METRICS_JSON:
{json.dumps({"avg_step_sec":round(avg,4),"min_step_sec":round(mn,4),"max_step_sec":round(mx,4),"peak_cuda_mem_gb":round(peak_mem_gb,2)})}')
            print('='*60 + '\n')
            sys.stdout.flush()
class MockState:
    global_step = 0
class MockArgs: pass
class MockControl: pass
gpu_name, total_mem = log_gpu_info()

# Синтетический датасет для классификации
N_SAMPLES = 100000
INPUT_DIM = 256
HIDDEN_DIM = 4096 # Большие слои для нагрузки на память
NUM_CLASSES = 10
BATCH_SIZE = 16384 # большой батч для нагрузки на память
NUM_EPOCHS = 50
torch.manual_seed(42)
X = torch.randn(N_SAMPLES, INPUT_DIM).float()
y = torch.argmax(X[:, :NUM_CLASSES] + torch.randn(N_SAMPLES, NUM_CLASSES) * 0.3, dim=1)

```

```

class SyntheticDataset(torch.utils.data.Dataset):
    def __init__(self, X, y):
        self.X = X
        self.y = y
    def __getitem__(self, idx):
        return {'input_ids': self.X[idx], 'labels': self.y[idx]}
    def __len__(self): return len(self.X)
model = nn.Sequential(
    nn.Linear(INPUT_DIM, HIDDEN_DIM),
    nn.ReLU(),
    nn.Dropout(0.1),
    nn.Linear(HIDDEN_DIM, HIDDEN_DIM),
    nn.ReLU(),
    nn.Dropout(0.1),
    nn.Linear(HIDDEN_DIM, HIDDEN_DIM),
    nn.ReLU(),
    nn.Dropout(0.1),
    nn.Linear(HIDDEN_DIM, HIDDEN_DIM // 2),
    nn.ReLU(),
    nn.Linear(HIDDEN_DIM // 2, NUM_CLASSES)
).cuda()

total_params = sum(p.numel() for p in model.parameters())
print(f'Всего параметров: {total_params}')
trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)
print(f'Обучаемых параметров: {trainable_params}')
device = next(model.parameters()).device
print(device)
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
dataset = SyntheticDataset(X, y)
metrics = MetricsCallback()
state = MockState()
args = MockArgs()
control = MockControl()
print('Starting training...')
print('TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!')
sys.stdout.flush()
t0 = time.time()
for epoch in range(NUM_EPOCHS):
    model.train()
    for i in range(0, len(dataset), BATCH_SIZE):
        metrics.on_step_begin(args, state, control)

        end_idx = min(i + BATCH_SIZE, len(dataset))
        batch_X = dataset.X[i:end_idx].cuda()
        batch_y = dataset.y[i:end_idx].cuda()

        optimizer.zero_grad()
        output = model(batch_X)
        loss = criterion(output, batch_y)
        loss.backward()
        optimizer.step()

        state.global_step += 1
        metrics.on_step_end(args, state, control)

    if epoch % 5 == 0:
        torch.cuda.empty_cache()
        gc.collect()
    metrics.on_train_end(args, state, control)
print(f'Done in {time.time()-t0:.2f} sec')
"
env:
- name: HF_TOKEN
  valueFrom:

```



```
secretKeyRef:
  name: hf-token-secret
  key: HF_TOKEN
  optional: true
resources:
  limits:
    cpu: "1"
    memory: 2Gi
    nvidia.com/gpu.shared: "1"
  requests:
    cpu: "1"
    memory: 1Gi
    nvidia.com/gpu.shared: "1"
```

Логи исполнения:

EXCLUSIVE

```
devchatay@epds73cmval2ppf6qhpm:~$ kubectl logs -n gpu-lab -f job/torch-training-exclusive
Installing dependencies...
WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by
'ReadTimeoutError("HTTPSConnectionPool(host='pypi.org', port=443): Read timed out. (read timeout=15)")': /simple/transformers/
WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by
'ReadTimeoutError("HTTPSConnectionPool(host='files.pythonhosted.org', port=443): Read timed out. (read timeout=15)")':
/packages/09/c8/844d5518a6ae4ffdc0cf0cae65ae13dbe5838306728c5c640b5a6e2a0c9/transformers-4.40.0-py3-none-any.whl.metadata
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager,
possibly rendering your system unusable.It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv. Use the
--root-user-action option if you know what you are doing and want to suppress this warning.
Starting training...
GPU: Tesla T4
TOTAL_GPU_MEMORY_GB: 14.56
cuda:0
Всего параметров: 43026442
Обучаемых параметров: 43026442
Starting training...
TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!
Step 2: time=0.010s, peak_mem=2.63GB
Step 4: time=0.923s, peak_mem=2.63GB
Step 6: time=0.006s, peak_mem=2.63GB
Step 8: time=0.006s, peak_mem=2.63GB
Step 10: time=0.877s, peak_mem=2.63GB
....
Step 338: time=0.991s, peak_mem=2.63GB
Step 340: time=0.988s, peak_mem=2.63GB
Step 342: time=0.999s, peak_mem=2.63GB
Step 344: time=0.111s, peak_mem=2.63GB
Step 346: time=0.006s, peak_mem=2.63GB
Step 348: time=0.988s, peak_mem=2.63GB
Step 350: time=0.999s, peak_mem=2.63GB
=====
avg_step_sec: 0.6763
min_step_sec: 0.0027
max_step_sec: 1.0909
peak_cuda_mem_gb: 2.63
METRICS_JSON: {"avg_step_sec": 0.6763, "min_step_sec": 0.0027, "max_step_sec": 1.0909, "peak_cuda_mem_gb": 2.63}
=====
Done in 303.72 sec
devchatay@epds73cmval2ppf6qhpm:~$
```

SHARED

```
devchatay@epds73cmval2ppf6qhpm:~$ for i in 1 2 3 4; do echo "===== torch-benchmark-ts-$i ====="; kubectl logs -n gpu-lab "job/torch-training-ts-$i"; done
```

```
===== torch-benchmark-ts-1 =====
```

Installing dependencies...

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>. Use the --root-user-action option if you know what you are doing and want to suppress this warning.

Starting training...

GPU: Tesla T4

TOTAL_GPU_MEMORY_GB: 14.56

Всего параметров: 43026442

Обучаемых параметров: 43026442

cuda:0

Starting training...

TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!

Step 2: time=0.018s, peak_mem=2.63GB

Step 4: time=1.966s, peak_mem=2.63GB

Step 6: time=0.006s, peak_mem=2.63GB

Step 8: time=0.006s, peak_mem=2.63GB

Step 10: time=2.508s, peak_mem=2.63GB

....

Step 342: time=4.257s, peak_mem=2.63GB

Step 344: time=0.530s, peak_mem=2.63GB

Step 346: time=0.018s, peak_mem=2.63GB

Step 348: time=4.274s, peak_mem=2.63GB

Step 350: time=4.212s, peak_mem=2.63GB

avg_step_sec: 2.9156

min_step_sec: 0.0044

max_step_sec: 4.3580

peak_cuda_mem_gb: 2.63

METRICS_JSON: {"avg_step_sec": 2.9156, "min_step_sec": 0.0044, "max_step_sec": 4.358, "peak_cuda_mem_gb": 2.63}

Done in 1289.25 sec

```
===== torch-benchmark-ts-2 =====
```

Installing dependencies...

WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ReadTimeoutError("HTTPSConnectionPool(host='files.pythonhosted.org', port=443): Read timed out. (read timeout=15)")':

/packages/09/c8/844d518a6aeb4ffdc0cf0cae65ae13dbe5838306728c5c640b5a6e2a0c9/transformers-4.40.0-py3-none-any.whl.metadata

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>. Use the --root-user-action option if you know what you are doing and want to suppress this warning.

Starting training...

GPU: Tesla T4

TOTAL_GPU_MEMORY_GB: 14.56

Всего параметров: 43026442

Обучаемых параметров: 43026442

cuda:0

Starting training...

TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!

Step 2: time=0.031s, peak_mem=2.63GB

Step 4: time=3.932s, peak_mem=2.63GB

Step 6: time=0.007s, peak_mem=2.63GB

Step 8: time=0.006s, peak_mem=2.63GB

Step 10: time=3.941s, peak_mem=2.63GB

....

Step 342: time=2.793s, peak_mem=2.63GB

Step 344: time=0.253s, peak_mem=2.63GB

Step 346: time=0.017s, peak_mem=2.63GB

Step 348: time=2.147s, peak_mem=2.63GB

Step 350: time=2.170s, peak_mem=2.63GB

avg_step_sec: 2.9169

min_step_sec: 0.0050

max_step_sec: 4.3611

peak_cuda_mem_gb: 2.63

METRICS_JSON: {"avg_step_sec": 2.9169, "min_step_sec": 0.005, "max_step_sec": 4.3611, "peak_cuda_mem_gb": 2.63}

Done in 1291.02 sec

===== torch-benchmark-ts-3 =====

Installing dependencies...

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>. Use the --root-user-action option if you know what you are doing and want to suppress this warning.

Starting training...

GPU: Tesla T4

TOTAL_GPU_MEMORY_GB: 14.56

Всего параметров: 43026442

Обучаемых параметров: 43026442

cuda:0

Starting training...

TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!

Step 2: time=0.010s, peak_mem=2.63GB

Step 4: time=1.961s, peak_mem=2.63GB

Step 6: time=0.008s, peak_mem=2.63GB

Step 8: time=0.018s, peak_mem=2.63GB

Step 10: time=2.511s, peak_mem=2.63GB

....

Step 340: time=4.193s, peak_mem=2.63GB

Step 342: time=4.271s, peak_mem=2.63GB

Step 344: time=0.524s, peak_mem=2.63GB

Step 346: time=0.016s, peak_mem=2.63GB

Step 348: time=4.280s, peak_mem=2.63GB

Step 350: time=4.211s, peak_mem=2.63GB

avg_step_sec: 2.9172

min_step_sec: 0.0037

max_step_sec: 4.3593

peak_cuda_mem_gb: 2.63

METRICS_JSON: {"avg_step_sec": 2.9172, "min_step_sec": 0.0037, "max_step_sec": 4.3593, "peak_cuda_mem_gb": 2.63}

Done in 1289.31 sec

===== torch-benchmark-ts-4 =====

Installing dependencies...

WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ReadTimeoutError("HTTPSConnectionPool(host='files.pythonhosted.org', port=443): Read timed out. (read timeout=15)")': /packages/09/c8/844d518a6aeb4ffdc0cf0cae65ae13dbe5838306728c5c640b5a6e2a0c9/transformers-4.40.0-py3-none-any.whl.metadata

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>. Use the --root-user-action option if you know what you are doing and want to suppress this warning.

Starting training...

GPU: Tesla T4

TOTAL_GPU_MEMORY_GB: 14.56

Всего параметров: 43026442

Обучаемых параметров: 43026442

cuda:0

Starting training...

TRAINING LOOP STARTED — GPU LOAD EXPECTED NOW!

Step 2: time=0.032s, peak_mem=2.63GB

Step 4: time=3.904s, peak_mem=2.63GB

Step 6: time=0.006s, peak_mem=2.63GB

Step 8: time=0.007s, peak_mem=2.63GB

Step 10: time=3.913s, peak_mem=2.63GB

....

Step 342: time=2.813s, peak_mem=2.63GB

```
Step 344: time=0.253s, peak_mem=2.63GB
Step 346: time=0.016s, peak_mem=2.63GB
Step 348: time=2.152s, peak_mem=2.63GB
Step 350: time=2.167s, peak_mem=2.63GB
```

```
=====
avg_step_sec: 2.9164
min_step_sec: 0.0039
max_step_sec: 4.3557
peak_cuda_mem_gb: 2.63
METRICS_JSON: {"avg_step_sec": 2.9164, "min_step_sec": 0.0039, "max_step_sec": 4.3557, "peak_cuda_mem_gb": 2.63}
=====
```

```
Done in 1290.70 sec
```

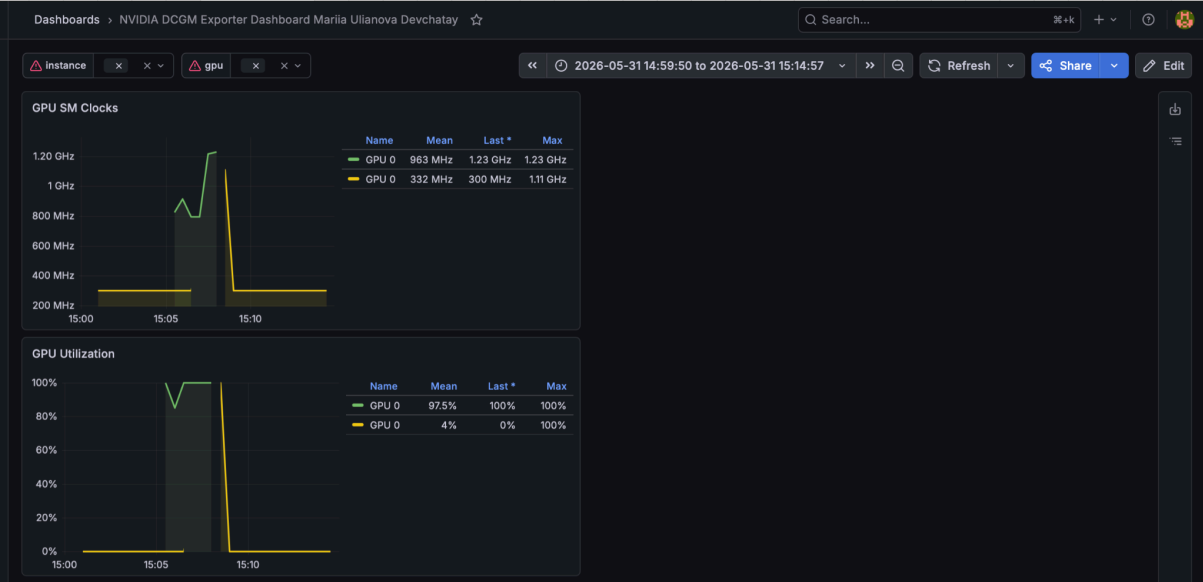
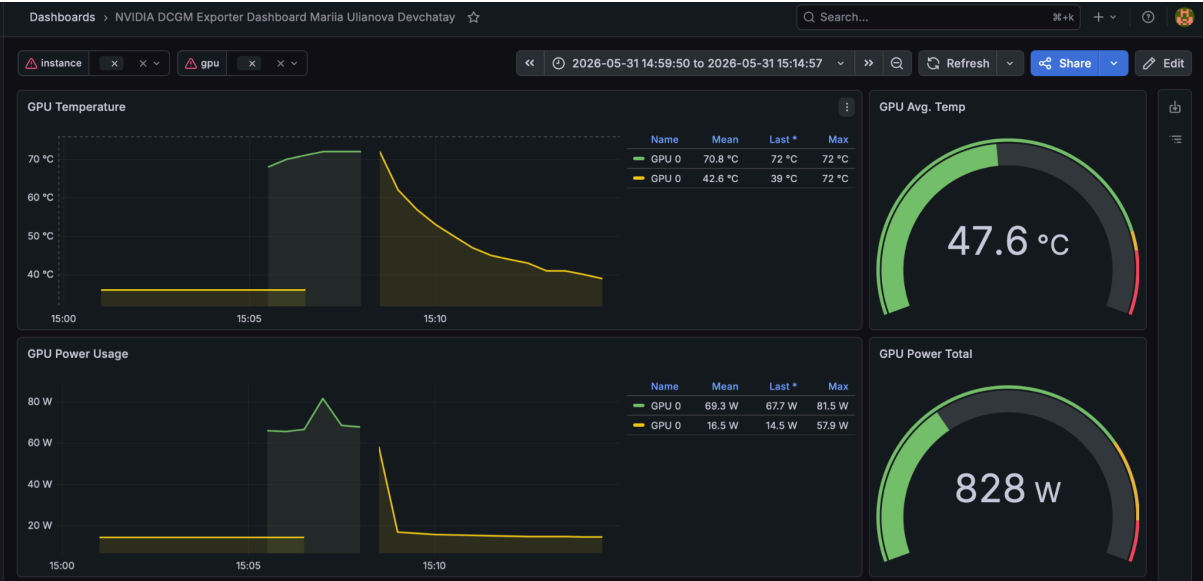
```
devchatay@epds73cmval2ppf6qhpm:~$
```

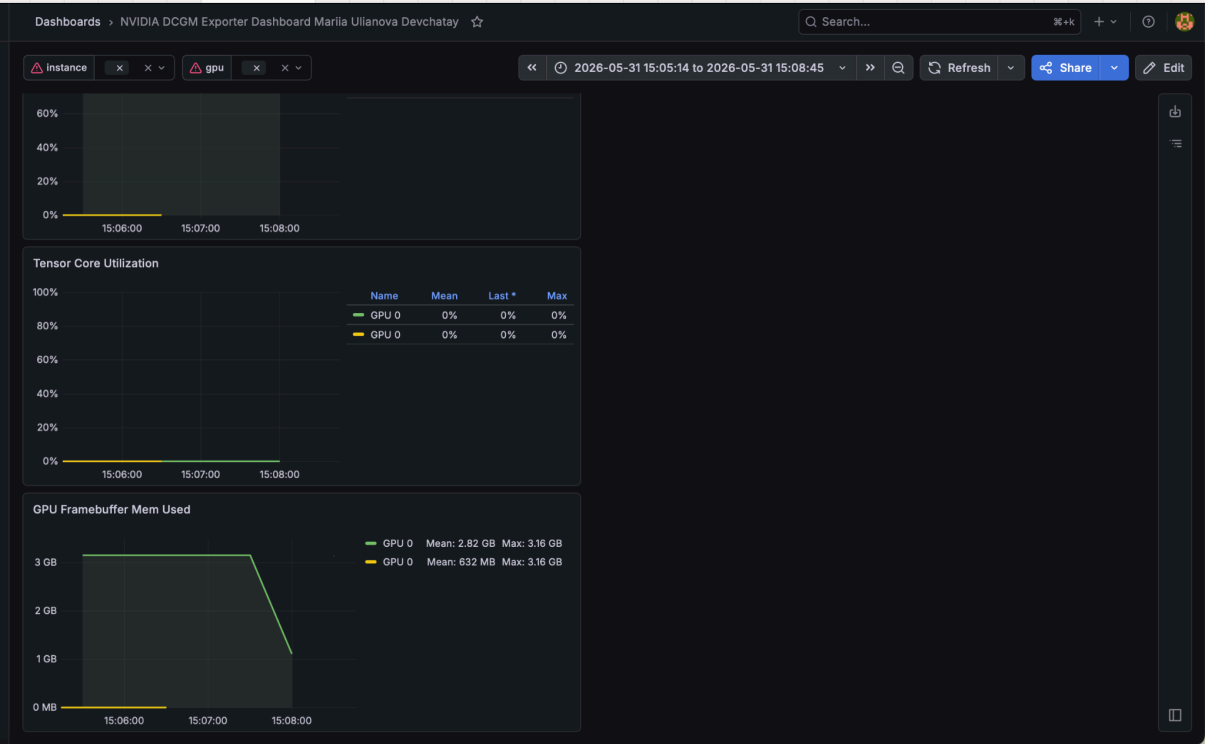
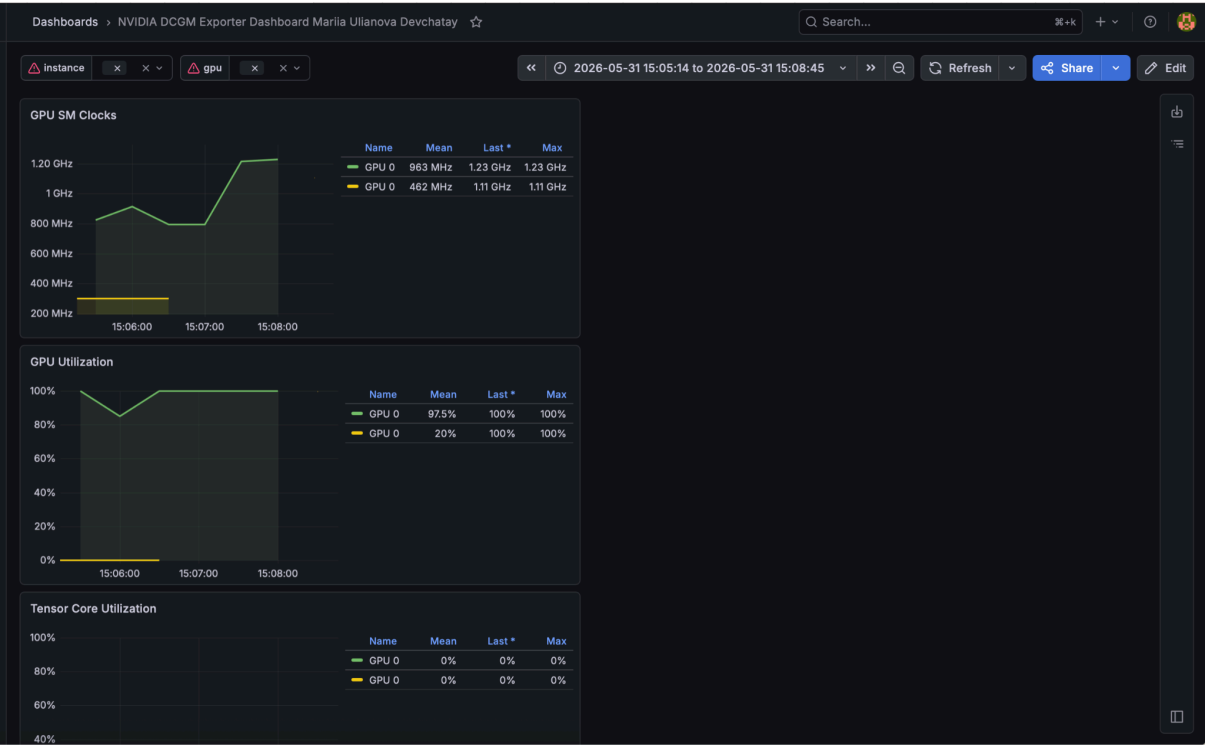
7. Графики или значения из Grafana по GPU utilization, memory usage и power usage;

EXCLUSIVE

На этой серии скриншотов показываю, что сначала GPU в ненагруженном состоянии находится, потом нагружается и, когда задача завершается, освобождается. Тут у меня возникли технические неполадки, я не разобралась, почему в каких-то случаях график по GPU 0 множится на несколько, хотя процесс был запущен ровно один.

Видно, что достаточно быстро исполнилась задача, температура поднялась с изначальных 36-39 до 47.6, мощность gpu power total небольшая, по памяти используем как раз ровно на одно обучение (2.8-3.6 ГБ).

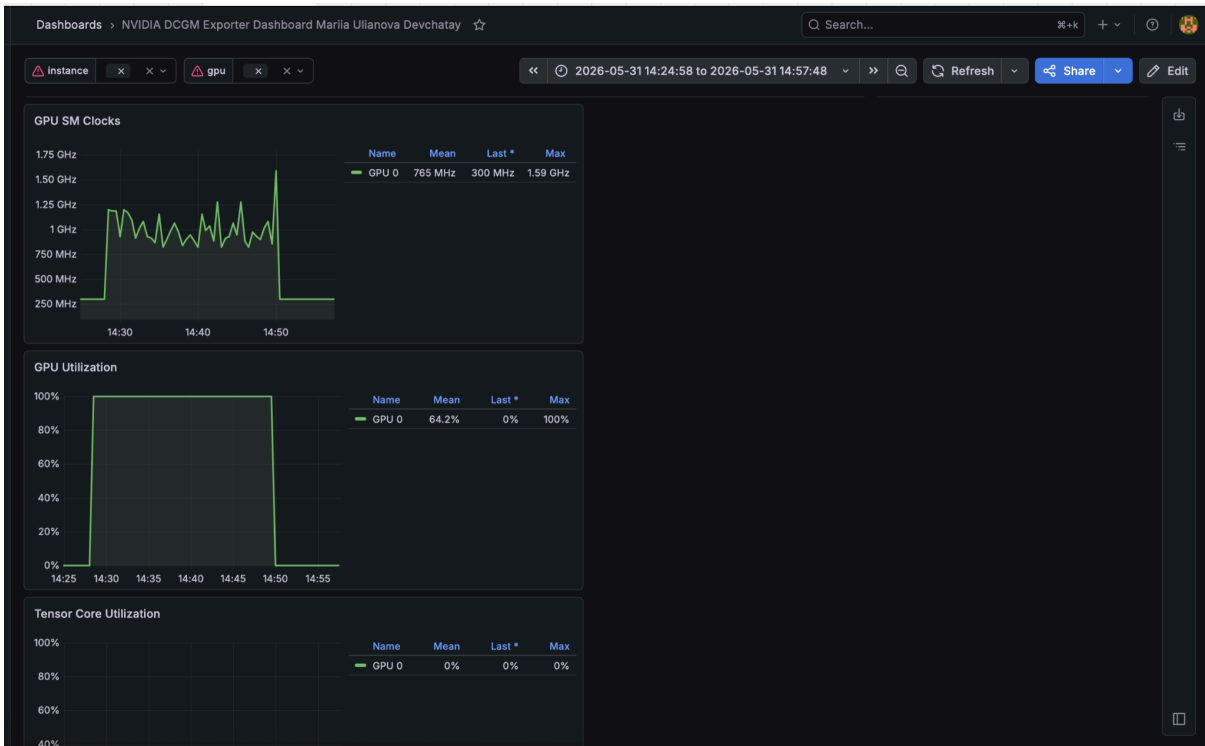
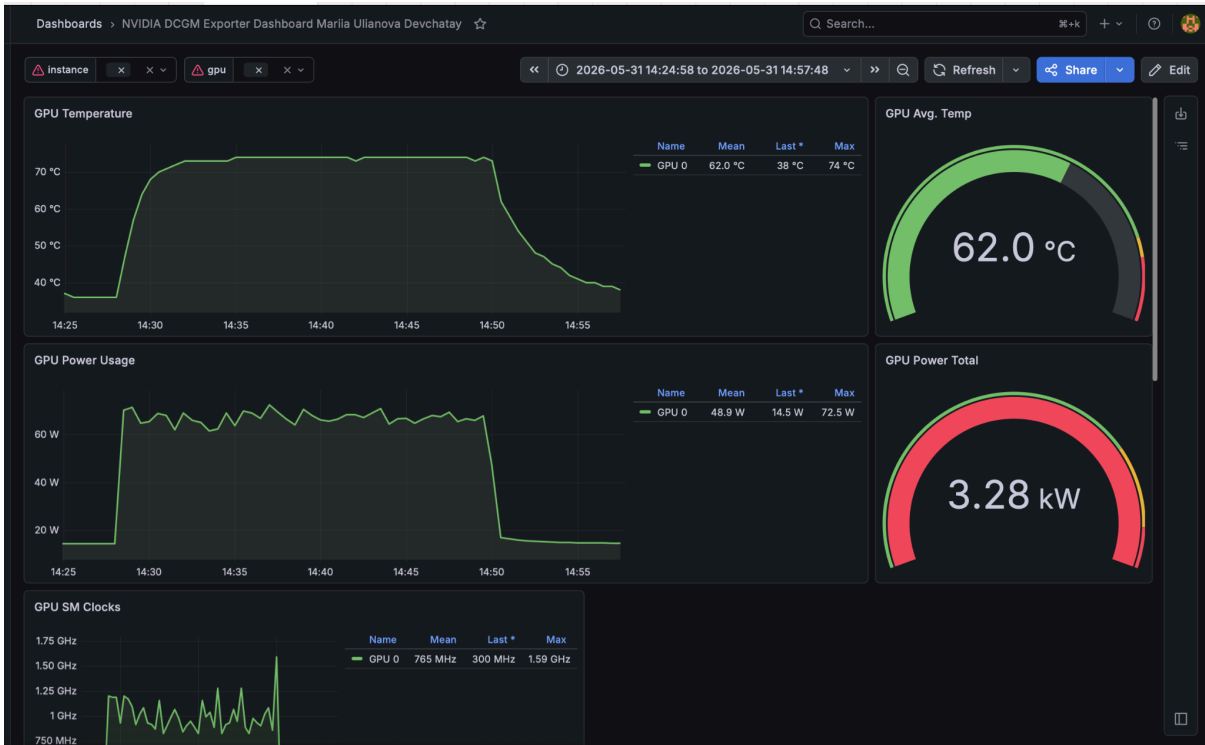


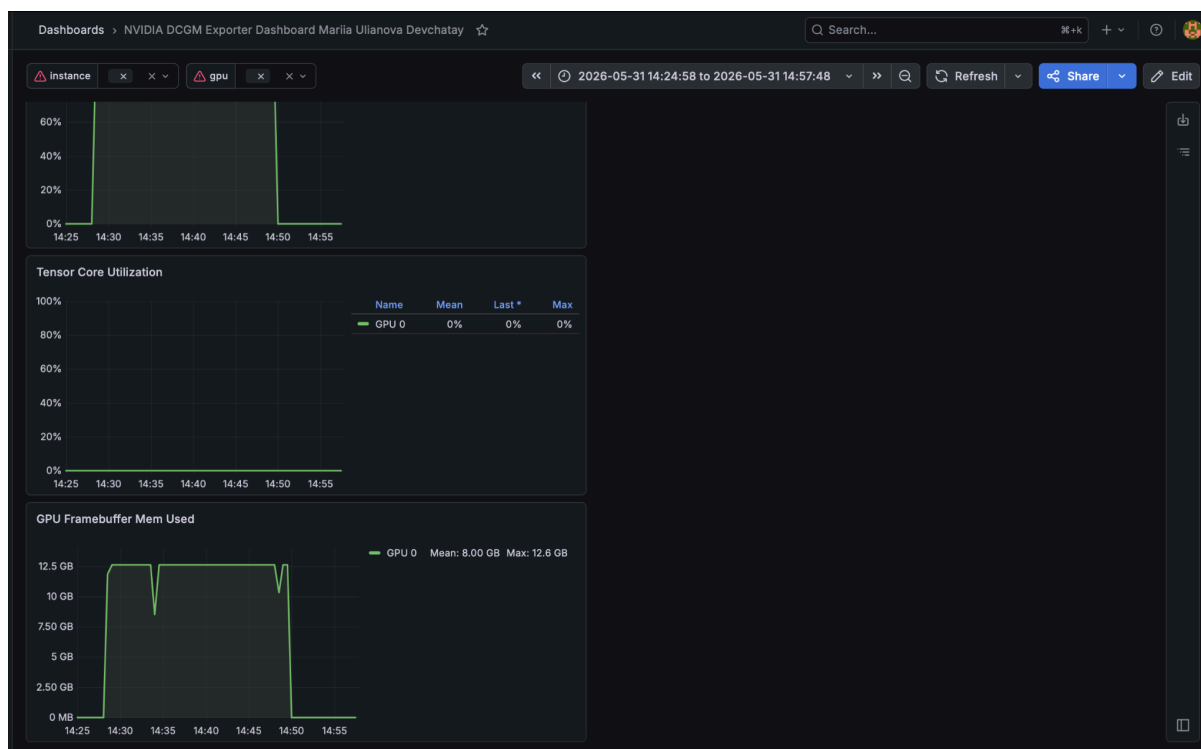


SHARED

Теперь по сравнению с shared. Было 4 пода с одной и той же задачей параллельно. Температура такая же, gpu power total достигла своего

критического значения. Видно, что по времени исполнялось сильно дольше, Exclusive 15.05-15.07, тут ~14.30-14.50, 2-3 минуты и 20 минут. Памяти использовалось тоже в 4 раза больше, максимально достигло 12 ГБ.





8. Вывод: какой режим подходит для одиночной тяжелой задачи, а какой для нескольких параллельных задач

Exclusive - одиночная тяжелая задача, так как все ресурсы - видеопамять, мощность и др. направлены только на одну задачу.

Shared - для нескольких параллельных задач. По результатам бенчмарка - время возросло в 4-6 раз, нагруженная тяжелая задача выполнялась параллельно в четырех подах. Уже на такой небольшой (43 млн параметров) сети достигли почти лимита по памяти.

Дополнительно стоит отметить поведение частот SM-ядер. В exclusive частота стабилизируется в районе 1.23 ГГц, что говорит о ровной нагрузке и отсутствии резких переключений контекста. В shared график SM Clocks демонстрирует более выраженные колебания и периодические просадки, что связано с накладными расходами планировщика time-slicing на переключение между четырьмя подами. Эти микропаузы накапливаются и объясняют, почему среднее время шага выросло не ровно в 4 раза. Также обращает на себя внимание

нулевая загрузка Tensor Core - это ожидаемое поведение, так как архитектура модели состоит из стандартных линейных слоёв FP32 и не использует операции смешанной точности, для которых предназначены тензорные ядра